FIITJEE

JEE (Advanced)-2018

ALL INDIA TEST SERIES

FULL TEST – XI

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 183**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01– 07, 19 – 25, 37 - 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (08 – 13, 26 – 31, 44 - 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (14 – 18, 32 – 36, 50 - 54)*** *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

1. Potential energy of a particle of mass m in xy plane is expressed by:-

$$U = \frac{(32/\sqrt{2})^2}{(x^2+y^2)^{\frac{3}{2}}}$$

Where x and y are the position coordinates. If at a point defined by position vector $\vec{r} = 2(\hat{i}+\hat{j})$. The particle is observed to be moving perpendicular to $\vec{r}$. Then which of the following is/are correct.

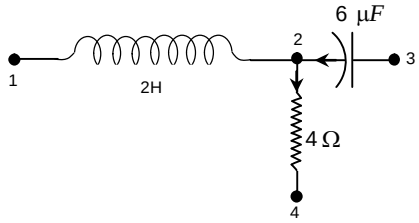
(A) $|\vec{a}_r|$ (Radial Acceleration) of the particle is $\frac{2\sqrt{2}}{m}\ m/s^2$

(B) $|\vec{a}_t|$ (Tangential Acceleration) of the particle is $\frac{\sqrt{2}}{m}\ m/s^2$

(C) Net force on the particle is $(\hat{i}+3\hat{j})N$

(D) Net force on the particle is $(2\hat{i}+3\hat{j})N$

2. The figure given is a part of bigger network. At the shown moment, the capacity of the capacitor is $6\,\mu F$ and decreasing at a constant rate $0.5\,\mu F/s$. The potential difference (V) across the capacitor at this moment is changing as under (Direction of current is shown in figure.)



$\frac{dV}{dt} = 2(volt/s)$; $\frac{d^2V}{dt^2} = \frac{1}{2}(volt/s^2)$

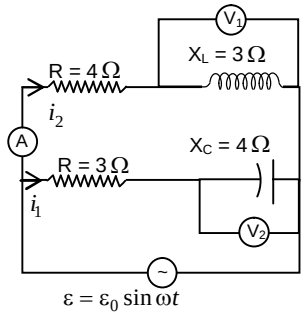
The current in the $4\,\Omega$ resistor is decreasing at the rate of $1\,\mu A/s$. The potential difference across the inductor at the moment is:-

(A) $1\,\mu V$ (B) $2\,\mu V$

(C) $3\,\mu V$ (D) $4\,\mu V$

Space for rough work

3. Consider the AC circuit shown and mark the correct option/s.



$\varepsilon = \varepsilon_0 \sin \omega t$

$[\varepsilon_0 = 20]$

(A) Phase difference between i_1 and i_2 is $\frac{\pi}{2}$

(B) Reading of voltmeter V_1 is $6\sqrt{2}V$

(C) Reading of voltmeter V_2 is $-8\sqrt{2}V$.

(D) Reading of ammeter is $2\sqrt{2}A$.

4. Two equal charges Q are at opposite corners of a square of side 'a' and an electric dipole of moment p is at a third corner, pointing towards one of the charges. If $p = 2\sqrt{2}a\ Q$; The field strength at the fourth corner of the square is:-

(A) $\frac{\sqrt{17/2}}{4\pi \in_0 a^2} Q$ (B) $\frac{\sqrt{17/4}}{4\pi \in_0 a^2} Q$

(C) $\frac{Q}{4\sqrt{2}\pi \in_0 a^2}$ (D) $\frac{\sqrt{3}}{4\pi \in_0 a^2} Q$

5. A radioactive sample is undergoing $\alpha -$ decay. At any time t_1 its activity is A and at a later instant t_2 it is $\frac{2A}{3}$. Which of the following is/are correct

(A) Mean life for the sample = $\frac{t_2 - t_1}{\log_e\left(\frac{3}{2}\right)}$

(B) Half life for the sample is $(t_2 - t_1)\left(\frac{\log_e 2}{\log_e \frac{3}{2}}\right)$

(C) Probability of survival of a nucleus after one mean life is 1/e.

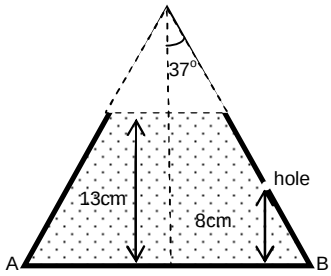
(D) Sum of lives of all nuclei is finite.

Space for rough work

6. A steel wire is stretched between two clamps 100 cm apart with a tension, which produces an extension of 0.608 cm in it. (Density of steel = 7600 kg/m^3; Young's modulus of elasticity = $2\times10^{11} N/m^2$)

(A) The fundamental frequency of its transverse vibration is 200 hz.
(B) The fundamental frequency of its transverse vibration is 400 hz.
(C) The 3rd overtone of the string will correspond to the frequency of 800 Hz.
(D) The 4th overtone of the string will correspond to the frequency of 600 Hz.

7. Curved surface of a vessel has shape of a truncated cone having semi vertex angle 37°. Vessel is full of water (density $\rho = 1000 kg/m^3$) upto a height of 13 cm and is placed on a smooth horizontal plane. Upper surface is opened to atmosphere. A hole of 1.5 cm^2 is made on curved wall at a height of 8 cm from bottom as shown in the figure. Area of water surface in the vessel is large as compared to the area of the hole



(A) Initial velocity of efflux is 1 m/s
(B) Initial horizontal range of water jet from point B is 6.65 cm
(C) Horizontal force required to keep the vessel in static equilibrium is 0.15 N.
(D) Horizontal force required to keep the vessel in static equilibrium is 0.12 N.

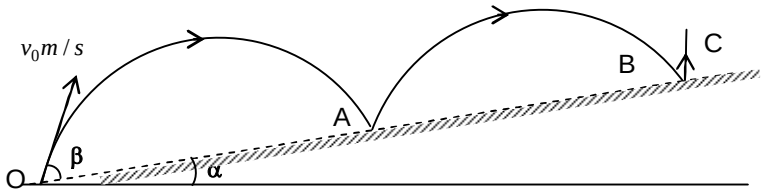
Space for rough work

Matrix Match Types

Each of 2 tables with 3 Columns and 4 Rows has three questions. Column-1 will be with 4 rows designated (I), (II), (III) and (IV). Column-2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column-3 will be with 4 rows designated (P), (Q), (R) and (S). Each question has four options with only **ONE** correct

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

A particle is projected along an inclined plane having inclination α, at angle β with the inclined plane. If coefficient of restitution is $\frac{1}{2}$, and it is given that the initial speed of particle is $v_0 = 10 m/s$, and that the particle bounces vertically upwards after 2 collisions as shown in figure, then answer the questions accordingly. (T denotes total time of flight and $K = \frac{T}{\sin\beta}$); (All units are in S.I. system)



Column – 1	Column – 2	Column – 3
(I) $\alpha = 45^o$	(i) $\beta = \cot^{-1}\left[\frac{13\sqrt{3}}{4}\right]$	(P) $K = \frac{13}{2\sqrt{2}} s$
(II) $\alpha = 30^o$	(ii) $\beta = \cot^{-1}\left[\frac{13\sqrt{3}}{12}\right]$	(Q) $K = 5.42s$
(III) $\alpha = 60^o$	(iii) $\beta = \cot^{-1}\left[\frac{13}{3}\right]$	(R) $K = \frac{13}{2\sqrt{3}} s$
(IV) $\alpha = 53^o$	(iv) $\beta = \cot^{-1}\left[\frac{13}{4}\right]$	(S) $K = 6.5s$

8. (I) (iv) (R) (II) (iii) (Q) (III) (i) (S) (IV) (iii) (Q)
How many of the above combinations are correct?
(A) 1 (B) 0
(C) 2 (D) 3

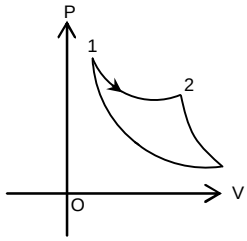
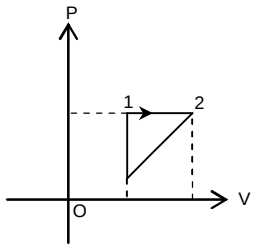
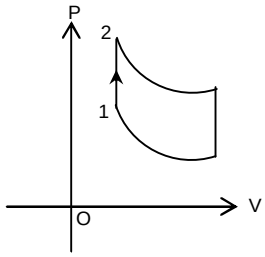
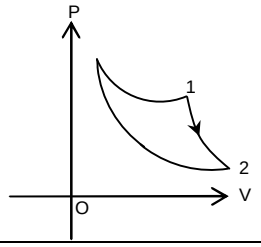
9. For which of the following combination will $T_{AB} = \frac{4\sqrt{2}}{\sqrt{185}}$ where T_{AB} is the time taken for particle to move from A to B.
(A) (I) (iv) (P) (B) (I) (iii) (R)
(C) (II) (iii (Q) (D) (IV) (iii) (Q)

10. In which of the case, will the velocity perpendicular to the plane after second collision be $\frac{7.5}{\sqrt{178}}$
(A) (I) (ii) (S) (B) (II) (i) (R)
(C) (III) (iv) (P) (D) (IV) (iii) (Q)

Space for rough work

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

An ideal monatomic gas (2 moles) undergoing a cyclic thermodynamic process in different ways as shown in corresponding diagrams in columns (3) of table. Consider only path from state 1 to state (2) Q; W; U denotes the correspoing heat exchange, workdone and internal energy for the system. For all shown process $1 \to 2$; pressure & volume is represented by $P_1; V_1$.

Column – 1	Column – 2		Column – 3
(I) $Q_{12} = 0$	(i) Isobaric	(P)	
(II) $\Delta U_{12} = 0$	(ii) Adiabatic	(Q)	
(III) $W_{12} = P_1(V_2 - V_1)$	(iii) Isothermal	(R)	
(IV) $Q_{12} = \frac{3(P_1V_1 - P_2V_2)}{2}$	(iv) Isochoric	(S)	

11. Which one of the following options is the correct combination?
(A) (II) (ii) (S) (B) (I) (ii) (S)
(C) (II) (i) (P) (D) (IV) (ii) (R)

Space for rough work

12. For which combination $\Delta Q = W$?

(A) (II) (iii) (P) (B) (III) (i) (P)
(C) (II) (i) (R) (D) (I) (i) (P)

13. For which one combination $\Delta Q = (5R)(\Delta T)$?

(A) (I) (iii) (P) (B) (II) (ii) (S)
(C) (III) (i) (Q) (D) (IV) (i) (Q)

SECTION – C

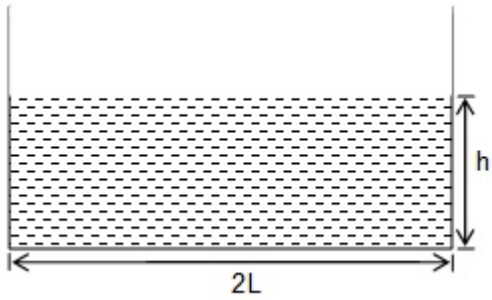
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

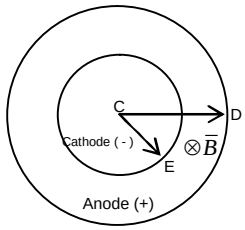
14. A rectangular pan of base (2L) is partly filled with water upto a height h as shown in figure. When the pan is pull a little, the water begins to slosh. Assuming that the water surface remains practically flat during sloshing the time period of the sloshing mode is given by:-

$$T = 2\pi\sqrt{\frac{L^2}{ngh}}$$

Find the value of n.



15. The electrodes of a diode are coaxial cylinders of radii inner 2m and outer 4 m. A potential difference V is maintained between them and their common axis is parallel to a uniform magnetic field $B = 1$ Tesla. The inner cylinder is the cathode, electrons leave it radially with negligible speed. If they will reach the anode at grazing incidence if $v = \frac{ne}{8m}$, then $n = ?$ ($CD = b$ and $CE = a$)

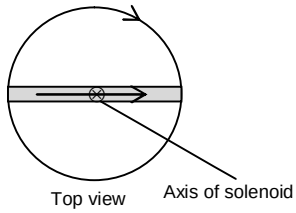


16. A homogeneous rod leans on a frictionless horizontal plane making an angle $\alpha = \pi/3$ with the horizontal. If released from rest it will fall. If the ratio of the length of the rod to the distance covered by the end on the plane at the beginning is given by n. Then n will be?

Space for rough work

17. A free electron with kinetic energy 3.4 eV happens to fall into an orbit around a fixed alpha particle forming a He^+ atom in the first excited state. A photon of wavelength λ_1 is emitted. Then the atom comes to the ground state emitting another photon of wavelength λ_2. If $\frac{\lambda_1}{\lambda_2} = \frac{12}{k}$, then find the value of k.

18. Circular coils are placed very close to each other so that they make up an infinite solenoid. The current in coils is 1A and number of coils/length of solenoid is 4000 turns/m. The diameter of the solenoid is 2 m. A wire of length equal to diameter of circular coil and carrying the same current is kept along the diameter as shown. Its mass is 4g. This wire is near the centre of solenoid. Find the velocity of the wire (in m/s) when it comes out of the solenoid. (Neglect friction and assume that rod is moving in the horizontal plane) (taken $\pi^2 = 10$) Neglect gravity.



Space for rough work

Chemistry

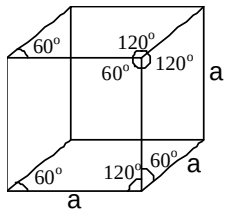
PART – II

SECTION – A

Straight Objective Type

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

19. A rhombohedral unit cell is shown What is its volume? Side length = $a\,\text{Å}$

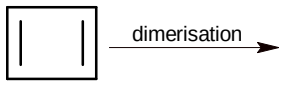


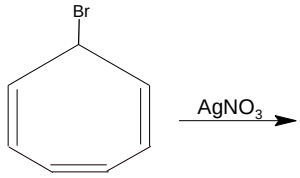
(A) $\frac{\sqrt{2}a^3}{2}$ (B) $\frac{a^3\sqrt{3}}{2}$

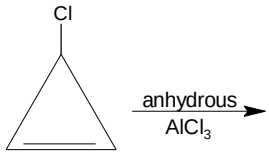
(C) $\frac{a^3}{\sqrt{2}}$ (D) $\sqrt[a^3]{\frac{3}{2}}$

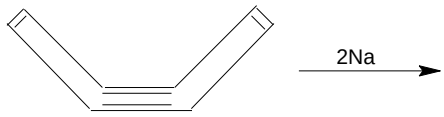
20. Which of the following pair/s have nearly same size?
(A) Zr, Hf (B) Nb,Ta
(C) Fe, Zn (D) K, Rb

21. Select the correct combination in which aromatic compound is/are formed in the reaction.

(A) 

(B) 

(C) 

(D) 

Space for rough work

22. For spontaneity select the correct statement/s
(A) $\partial G < O$ at constant, temperature and pressure for irreversible process.
(B) $\partial G < O$ at constant, temperature and volume for irreversible process.
(C) $dS > O$ at constant, internal energy and volume for an isothermal irreversible process.
(D) $dU < O$ at constant, entropy and volume for an isothermal irreversible process.

23. In a chemical reaction:

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$$

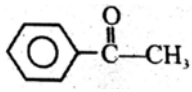
if pressure of the system get decrease then :

(A) Degree of dissociation of PCl_5 increases, net concentration of all decreases
(B) Degree of dissociation of PCl_5 decreases, net concentration of all increases
(C) Vapour density of the mixture decreases
(D) Vapour density of the mixture increases

24. Compound(A)$C_8H_8O \xrightarrow[HCl]{NH_2OH}$ B + C; C $\xrightarrow{H^+}$ D

B $\xrightarrow{H^+}$ E $\xrightarrow[(ii)\ H^+/H_2O]{(i)\text{Boil NaOH}}$ G($C_7H_6O_2$) (white solid)

D $\xrightarrow[(ii)\ H^+/H_2O]{(i)KOH}$ F(C_6H_7N) (Oil) $\xrightarrow[\text{rapid}]{CH_3COCl}$ D

[B, C, D and E are isomers having molecular formula C_8H_9NO]
Which of the following statement is/are correct regarding above reaction:
(A) Compound(A) is an aldehyde and can give (+ ve) Tollen's test
(B) Compound (B) and (C) can show geometrical isomerism
(C) Compound (A) is an aromatic ketone which having structure 
(D) Compound (G) is more acidic than CH_3COOH

25. Select the correct statement/s:
(A) Square planer complex can show enantiomerism
(B) In $KMnO_4$ central atom is sp^3 hybridised and its geometry is tetrahaedral
(C) $[PtF_6]^{2-}$ is inner orbital complex while $[Co(NH_3)_6]^{2+}$ is outer orbital complex
(D) The order of crystal field splitting energy in tetrahaedral, octahedral and square planar complex are $\Delta_t < \Delta_0 < \Delta_{sp}$ respectively

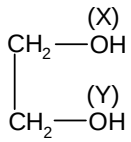
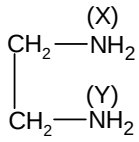
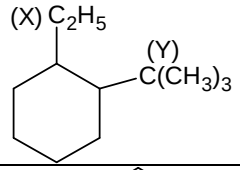
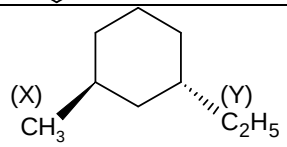
Space for rough work

Matrix Match Types

Each of 2 tables with 3 Columns and 4 Rows has three questions. Column-1 will be with 4 rows designated (I), (II), (III) and (IV). Column-2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column-3 will be with 4 rows designated (P), (Q), (R) and (S). Each question has four options with only **ONE** correct

Answer questions 26, 27 and 28 by appropriately matching the information given in the three columns of the following table.

If two different 3 – D arrangements in space of atom in a molecule are interconvertible merely by free rotation about σ bond is called conformations.

Column – 1		Column – 2	Column – 3
(I)		(i) Gauche conformer most stable	(P) Tortional & Vander Waal's strain is decrease.
(II)	 at pH = 4 in aq. solution	(ii) Anti/staggered conformer most stable	(Q) Intramolecular Hydrogen Bonding
(III)		(iii) group (X) is at axial position while group (Y) is at equatorial for stable conformer	(R) Trans form
(IV)		(iv) group (X) & (Y) both are at equatorial position for stable conformer	(S) Cis form

26. Select the correct one:
(A) (IV), (iv), (S) (B) (III), (iii), (S)
(C) (I), (iv), (Q) (D) (I), (ii), (P)

Space for rough work

27. Select the incorrect one:
(A) (I), (i), (Q) (B) (I), (ii), (P)
(C) (III), (iv), (R) (D) (IV), (iii), (P)

28. The correct one with respect to ethylene diamine.
(A) (II), (i), (P) (B) (II), (i), (Q)
(C) (II), (ii), (R) (D) (II), (ii), (Q)

Answer questions 29, 30 and 31 by appropriately matching the information given in the three columns of the following table.

Column – 1	Column – 2	Column – 3
(I) Very dilute HNO_3	(i) reacts with copper	(P) Metal nitrate + NO are formed
(II) Dilute HNO_3	(ii) reacts with Tin	(Q) Metal nitrate + NO_2 are formed
(III) Concentrated HNO_3	(iii) reacts with Zinc	(R) Metal nitrate + N_2O are formed
(IV) Aqua regia	(iv) reacts with Iron	(S) Metal nitrate + NH_4NO_3 are formed

29. From the given columns the correct set is:
(A) (I) (i) (Q) (B) (II) (i) (P)
(C) (IV) (i) (Q) (D) (III) (iv) (R)

30. From the given columns the incorrect is:
(A) (I) (ii) (S) (B) (III) (iii) (Q)
(C) (II) (ii) (P) (D) (II) (iv) (R)

31. HNO_3 reacts with metal (Tin) will give the correct reaction by the following combination____
(A) (III) (ii) (S) (B) (III) (ii) (R)
(C) (I) (ii) (P) (D) (II) (ii) (S)

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

32. The number of Hg – O bond present in chemical formula of iodide of million's base____

33. The total no. of amide/s possible with the molecular formula $C_5H_{11}NO$ is X. Find the value of $\frac{X}{4}$.

34. 3.838L water is added in 300 ml of 0.1M HA so that its $[H^+]$ is decreased by a factor of X Ka for HA = 1.6×10^{-4} Find the approximate value of X.

35. How many of them shows minimum boiling azeotrope;
Acetone and benzene
Acetone and carbon disulphide
Acetone and ethanol
Acetone and aniline
Carbon tetra chloride and silicon tetra chloride
Carbon tetra chloride and chloroform
Benzene and chloroform
Diethyl ether and chloroform
n – butyl chloride and n – butyl bromide

36. Oxidation number of Fe in the complex formed during NO_3^- ring test is 'a' and oxidation number of Cr in K_3CrO_8 is 'b' then 'a + b' is:

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

37. Let X be a set $\{1,2,3,4,5,.....,n\}$ and P the subset $\{1,2,3,4,5\}$. The number of subsets Q of X such that $n(P \cap Q) = \alpha$, is

(A) $2^n - 2^{n-5}$ if $\alpha < 5$ (B) $2^n - 2^{n-5}$ if $\alpha > 0$

(C) ${}^{n-5}C_2 . {}^5C_2$ if $\alpha = 2$ (D) ${}^5C_1 . \sum_{r=0}^{n-5} {}^{n-5}C_r$ if $\alpha = 1$

38. If $\int \frac{dx}{x^{1/3}+2} = g(x) + c$ where $g(0) = 12\ln 2$ then $2g(-1)$ is a multiple of

(A) 2 (B) 3

(C) 5 (D) 7

39. There are 2m people, m people wearing red shirts and m people wearing blue. The number of ways in which these 2m people can stand in a queue, such that at any instant a person standing in queue sees more number of red shirt people standing in front of him than the blue shirt people is

(A) 4 if m = 2 (B) 5 if m = 3

(C) 14 if m = 4 (D) 42 if m = 5

40. If A and B be two points on the curve $y = \log_{\frac{1}{3}}(x - 0.\overline{3}) + \log_3 \sqrt{9x^2 - 6x + 1}$ and A is also a point on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$. B lies inside the ellipse having integral co ordinates. Then

(A) $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3$ (B) $|\overrightarrow{AB}| = 1$

(C) $|\overrightarrow{AB}| = 1$ or $|\overrightarrow{AB}| = 3$ (D) $\Delta_{AOB} = 1$ sq. unit

Space for rough work

41. Consider the curves $C_1 \equiv x^2 + y^2 - 6x - 8y + 24 = 0$ and $C_2 \equiv 4x^2 + 5y^2 - 40x + 30y + 125 = 0$. The set of values of c for which $y = 2x + c$ does not intersect any of the curves and lies between them, is.

(A) $c \in (-\infty, -13 + 2\sqrt{6}) \cup (-2 - \sqrt{5}, \infty)$ (B) $c \in (-2\sqrt{6} - 13, \infty)$

(C) $c \in [2\sqrt{6} - 13, \sqrt{5} - 2]$ (D) $c \in (-13 + 2\sqrt{6}, -2 - \sqrt{5})$

42. Consider f in [0, 2] where

$f(x) = |4x - 5|[x]$ for $x > 1$

$[\cos \pi x]$ for $x \le 1$; where $[\cdot]$ denotes G. I. F.

The points of discontinuities of this function say $(\alpha, \beta, \gamma, \delta)$ in increasing order satisfy $M = \alpha \cos 3t + \beta \cos 2t + \gamma \cos t + \delta$ $(t \in R)$. Then

(A) $M_{\max} = \frac{9}{2}$ (B) $M_{\min} = \frac{5}{4}$

(C) $M_{\max} = \frac{7}{2}$ (D) $M_{\min} = \frac{3}{4}$

43. Which of the following is/are incorrect?

(A) The ratio $\frac{\Delta}{\Delta_E}$ is equal to $2 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$ (Where Δ_E denotes area of triangle formed by joining the excentres)

(B) The radius of escribed circles of ΔABC are r_1, r_2 and r_3 respectively (opposite to vertices A, B and C) It $r_1 + r_2 = 3R$ and $r_2 + r_3 = 2R$ (Where R is circum radius of ΔABC), then $\angle C = 90^\circ$.

(C) In a right angled triangle, acute angle α, β satisfy $\tan \alpha (1 + \tan \alpha) + \tan \beta (1 + \tan \beta) = 4$. If hypotenuse is of length d, then area of triangle is $\frac{d^2}{4}$

(D) None of these

Space for rough work

Matrix Match Types

Each of 2 tables with 3 Columns and 4 Rows has three questions. Column-1 will be with 4 rows designated (I), (II), (III) and (IV). Column-2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column-3 will be with 4 rows designated (P), (Q), (R) and (S). Each question has four options with only **ONE** correct

Answer questions 44, 45 and 46 by appropriately matching the information given in the three columns of the following table.

Column –1: Contains functions/expressions

Column –2 & 3: Contain properties that these functions/expressions exhibit

Column – 1	Column – 2	Column – 3
(I) Consider $f:R\to R$ $f(x)=2x+[x]+\sin x\cos x$	(i) One One	(P) Continuous
(II) Consider $f:[4,6]\to R$ $f(x)=\tan\left(2\sqrt{x-4}+\sqrt{6-x}\right)$	(ii) Many One	(Q) Differentiable
(III) Consider $f:[-1,1]\to\left[\frac{13}{36},\frac{19}{20}\right]$ $f(x)=\frac{1}{\pi}\left(\sin^{-1}x+\cos^{-1}x-\tan^{-1}x\right)+\frac{1}{x^2+2x+6}$	(iii) Into	(R) Not Differentiable at exactly one point
(IV) Let a,b,c be real numbers such that $3b>a^2$. Then the function $f:R\to R$ given by $f(x)=x^3+ax^2+bx+c$ is	(iv) Onto	(S) Discontinuous

44. Which of the following is a correct combination?
(A) (I), (iv), (S) (B) (IV), (iii), (Q)
(C) (II), (i), (R) (D) (III), (iv), (Q)

45. Which of the following is a correct combination?
(A) (I), (iii), (S) (B) (II), (i), (R)
(C) (III), (iii),(P) (D) (IV), (ii), (P)

46. Which of the following is a correct combination?
(A) (I), (ii), (P) (B) (I), (i), (S)
(C) (III), (ii), (P) (D) (IV), (ii), (Q)

Space for rough work

Answer questions 47, 48 and 49 by appropriately matching the information given in the three columns of the following table.

Column – 1	Column – 2	Column – 3
(I) Let $P(x)$ be a bi quadratic function of x such that $\lim_{x\to 0}\left[\frac{P(-x)}{2x^3}\right]^{\frac{1}{x}} = \frac{1}{e^3}$	(i) $P(x)$ has maxima but no minima	(P) $P(x)$ is continuous for all x in its domain
(II) Consider the functions $P(x)$ and $Q(x)$ such that $P(x) = \frac{x^3}{2} + 1 - x\int_0^x Q(t)dt.$ and $Q(x) = x - \int_0^1 P(t)dt$ Both $P(x)$ and $Q(x)$ are defined from $R \to R$	(ii) $P(x)$ has minima but no maxima	(Q) $P(x)$ is discontinuous at exactly 2 points
(III) Consider function $P : R \to R$ defined by $P(x) = \frac{x^2 - a}{x^2 + a}; 0 < a < 9$	(iii) $P(x)$ has both local minima and maxima	(R) $P(x)$ is bounded
(IV) consider $P(x) = xe^x + (xe^x)^{-1}$ $\forall x \in R - \{0\}$	(iv) $P(x)$ is decreasing in $(-\infty, 0)$ and increasing in $(0, \infty)$	(S) $P(x)$ attains all values from $(-\infty, \infty)$

47. Which of the following is the correct combination?
(A) (IV), (iv), (R) (B) (I), (ii), (Q)
(C) (III), (ii), (R) (D) (II), (i), (P)

Space for rough work

48. Which of the following is the correct combination?
(A) (II), (ii), (R) (B) (I), (i), (R)
(C) (IV), (ii), (P) (D) (III), (iv), (R)

49. Which of the following is the incorrect combination?
(A) (III), (ii), (P) (B) (II), (ii), (P)
(C) (III), (iv), (R) (D) (I), (ii), (P)

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

50. The number of factors of $2^{15}3^{10}5^{6}$ which are either perfect squares or perfect cubes (but not both) are α. Then $\left[\frac{\alpha}{42}\right]$ is equal to_____ [where $[\cdot]$ denotes G.I.F]

51. n^2+3n+6 is a perfect square is satisfied by a set of integers $\{a_1, a_2, ..., a_k\}$. Let $P(x)$ be a polynomial of least possible degree such that

$$\left.\begin{array}{ll} p(a_i)=(a_i)^2 & a_i<0 \\ P(a_i)=0 & a_i\geq 0 \end{array}\right\} \forall\ 1\leq i\leq k$$

If the sum of coefficients of $P(x)$ can be expressed as $\frac{m}{n}$ in the lowest form, then $|m+n|$ is equal to?

52. Let $f(x)=\frac{(2x^3-9x^2+12x+3)\ (x-\alpha)}{(x-\alpha)}$. If range of $f(x)$ is a proper subset of R, then $\alpha\in(-\infty,m)\cup(n,\infty)$. The value of $m+n$ equals_____.

53. If $\int_0^1 \frac{dx}{\sqrt{1+x}+\sqrt{1-x}+2}=\sqrt{a}-\sqrt{b}-\frac{\pi}{\sqrt{c}}$, where $a,b,c\in N$, then value of $\left[\frac{a^2+b^2+c^2}{10}\right]$ is equal to_____. [where $[\cdot]$ denotes G.I.F.]

54. Let $D=\{(x,y)\in R^2 : x^2+y^2\leq 1\}$. Then the maximum number of points in D such that distance between any pair of points is at least 1 will be_____.

Space for rough work